

AI-Assisted Triage Literature Review & Problem Definition

Paper 1: Da'Costa et al. (2025) Da'Costa, A., Teke, J., Origbo, J.E., Osonuga, A., Egbon, E., & Olawade, D.B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*, 197, 105838.

Problem: The clinical problem is ED overcrowding and inconsistent patient prioritization driven by subjective clinician judgement, which normally occurs during peak hours and mass casualty events.

Method: The method is a narrative review of peer-reviewed literature from 2015-2024, examining AI-driven triage technologies, their benefits, challenges, and future applications.

Outcome: AI integration minimized subjective human error, optimized bed/staff scheduling, and cut average patient wait times by up to 30% while reducing time-to-treatment by 20%, hence machine learning and natural language processing were identified as valuable technologies.

Limitation: Algorithmic bias across historical training datasets that can lead to inequitable prioritization in minority cohorts, alongside significant challenges regarding clinician trust in opaque "black-box" systems.

Paper 2: Tyler et al. (2024) Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D.W., Ginsberg, B., Ahmad, H., & Jacobs, R.J. (2024). Use of artificial intelligence in triage in hospital emergency departments: A scoping review. *Cureus*, 16(5), e59906.

Problem: The clinical problem is the inconsistency and subjectivity of standard five level triage scales, which depend heavily on nurse experience and are prone to under-triaging and over-triaging critically ill patients.

Method: The authors conducted a scoping review of published studies between 2013 and 2023. They searched multiple databases and analyzed 29 studies examining the use of AI and machine learning in emergency department triage.

Outcome: The review consistently finds ML models outperforming conventional triage systems, KATE™ achieved 75.7% accuracy versus 59.8% for nurses, and one mode reached an AUROC of 0.991 for predicting critical pediatric outcomes.

Limitations: The foundational studies heavily relied on retrospective, single-center data arrays with significant gaps in raw text preprocessing, creating localized selection biases.

Paper 3: Araouchi & Adda (2024) Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-assisted multimodal triage system for health centers. *Procedia Computer Science*, 251, 430–437.

Problem: Severe emergency department overcrowding and rising clinical workloads driving manual assessment delays, human errors, and high inter-rater variability under standard triage protocols.

Method: The method involves training and evaluating six supervised ML on 1,267 real patient records from two EDs, with SMOTE applied to address class imbalance, then combining the best three performers into a stacked ensemble.

Outcome: The final Stacking Model outperformed all individual standalone models, achieving a triage classification accuracy of 80.05% and an AUROC score of 0.93.

Limitation: The entire study relies on a single Kaggle-sourced dataset from two Korean EDs in 2016-2017, which severely restricts generalizability, the models have not been validated across different demographics, seasonal contexts, or hospital systems.

Draft Problem Statement

Many emergency department triage relies on subjective nurse assessment, producing inconsistent patient prioritisation and measurable rates of under triage in high acuity cases. Existing AI triage research demonstrates consistent ML model superiority over conventional systems, yet nearly all published studies use retrospective, single site data and none report prospective validation in a Caribbean or small island context. This twelve week pilot will develop and evaluate a machine learning decision support tool to improve acuity classification accuracy and reduce under triage rates among high acuity presentations.